(AWS Cafe Challenge Portfolio)

Challenge 5 - Creating a Scalable and Highly Available Environment for the Café

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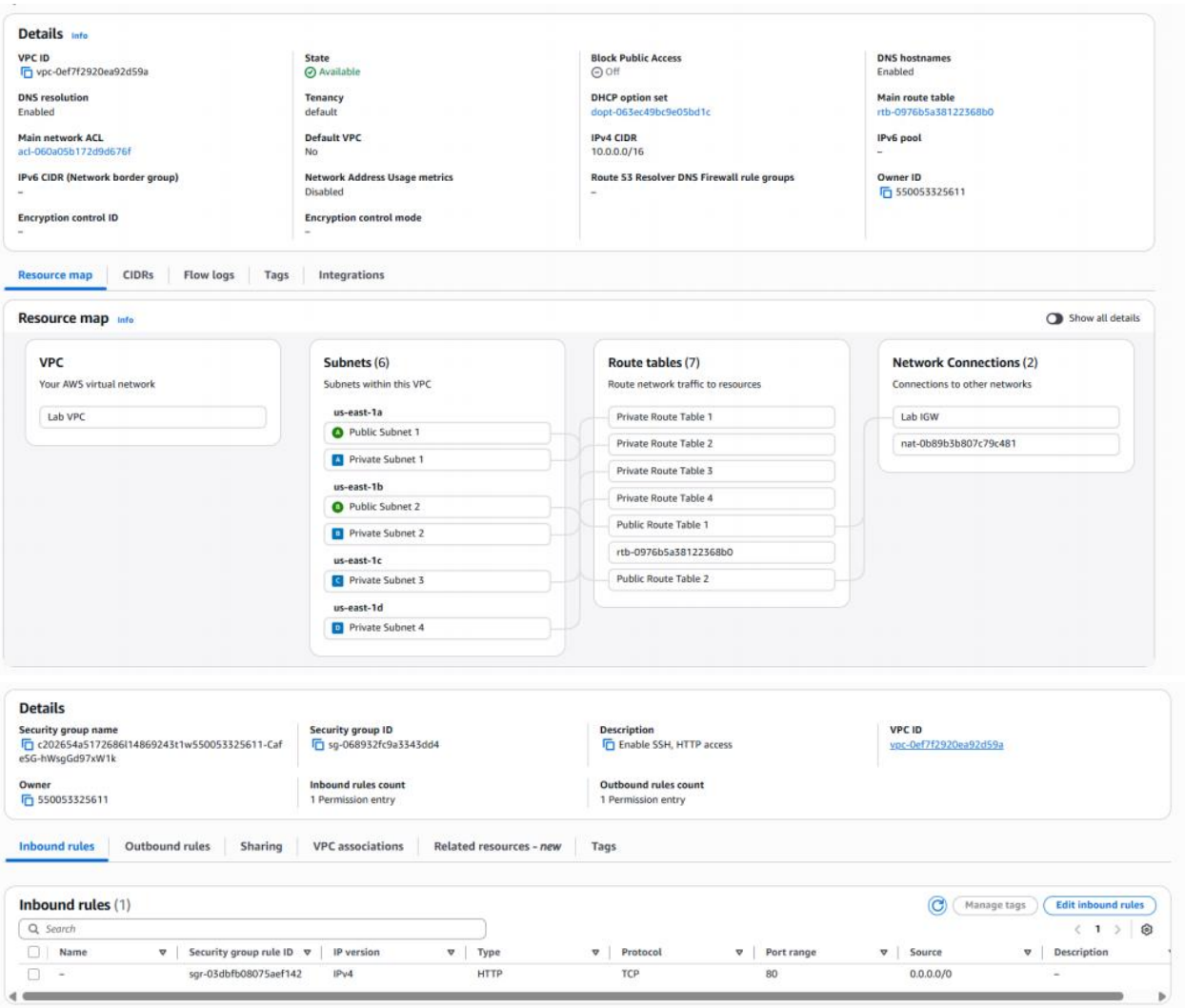
Please submit your workshop maintaining the following guidelines through the portal opened in MST

Challenge (Cafe) Lab: Creating a Scalable and Highly Available Environment for the Café

Task 1: Inspecting your environment

Show a screenshot of the Amazon VPC console displaying the Lab VPC details, including:

- The list of subnets (Public Subnet 1, Public Subnet 2, Private Subnet 1, Private Subnet 2) with their Availability Zones visible.
- The CafeSG security group inbound rules showing which ports are open.

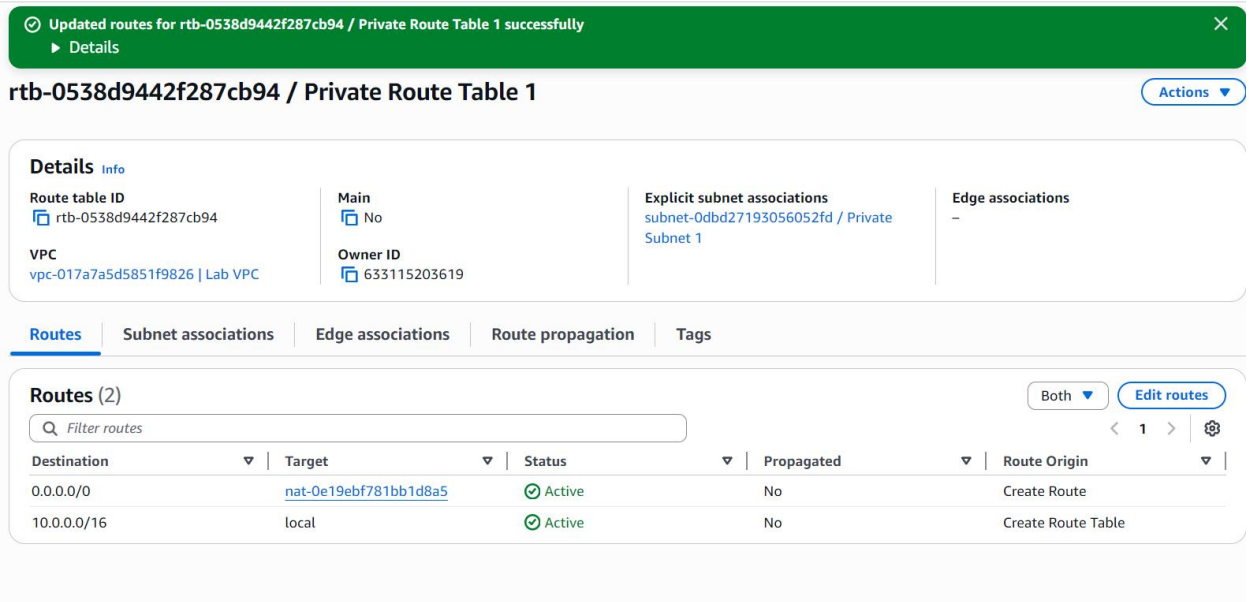

Details Info
VPC ID
vpc-0ef7f2920ea92d59a
State
Available
Block Public Access
Off
DNS hostnames
Enabled
DNS resolution
Enabled
Tenancy
default
DHCP option set
dopt-063ec49bc9e05bd1c
Main route table
rtb-0976b5a38122368b0
Main network ACL
acl-060a05b172d9d676f
Default VPC
No
IPv4 CIDR
10.0.0.0/16
IPv6 pool
-
IPv6 CIDR (Network border group)
-
Network Address Usage metrics
Disabled
Route 53 Resolver DNS Firewall rule groups
-
Owner ID
550053325611
Encryption control ID
-
Encryption control mode
-
Resource map
CIDRs
Flow logs
Tags
Integrations
Resource map Info
Show all details
VPC
Your AWS virtual network
Lab VPC
Subnets (6)
Subnets within this VPC
us-east-1a
Public Subnet 1
Private Subnet 1
us-east-1b
Public Subnet 2
Private Subnet 2
us-east-1c
Private Subnet 3
us-east-1d
Private Subnet 4
Route tables (7)
Route network traffic to resources
Private Route Table 1
Private Route Table 2
Private Route Table 3
Private Route Table 4
Public Route Table 1
rtb-0976b5a38122368b0
Public Route Table 2
Network Connections (2)
Connections to other networks
Lab IGW
nat-0b89b3b807c79c481

Details
Security group name
c202654a5172686l14869243t1w550053325611-CafeSG-hWsgGd97xW1k
Security group ID
sg-068932fc9a3343dd4
Description
Enable SSH, HTTP access
VPC ID
vpc-0ef7f2920ea92d59a
Owner
550053325611
Inbound rules count
1 Permission entry
Outbound rules count
1 Permission entry
Inbound rules
Outbound rules
Sharing
VPC associations
Related resources - new
Tags
Inbound rules (1)
Manage tags
Edit inbound rules
Search
Name
Security group rule ID
IP version
Type
Protocol
Port range
Source
Description
-
sgr-03dbfb08075aef142
IPv4
HTTP
TCP
80
0.0.0.0/0
-

Task 2: Updating a network to work across multiple Availability Zones

Show two screenshots:

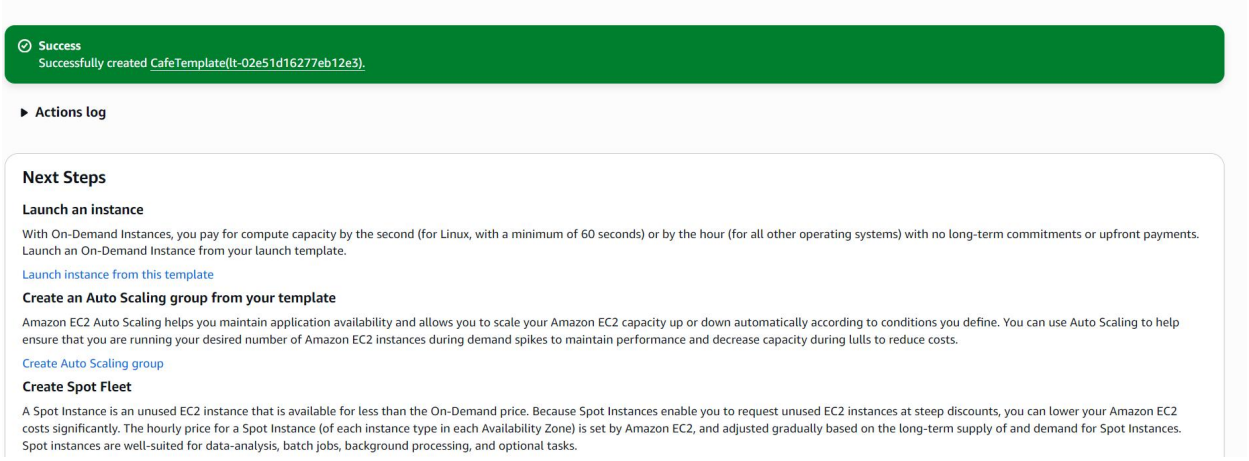
- The VPC console NAT Gateways list showing the newly created NAT gateway in Public Subnet 2 with status Available.
- The route table associated with Private Subnet 2, showing a route for 0.0.0.0/0 pointing to the new NAT gateway.



Task 3: Creating a launch template

Show a screenshot of the EC2 Launch Templates console displaying the details of your newly created launch template, with the following visible:

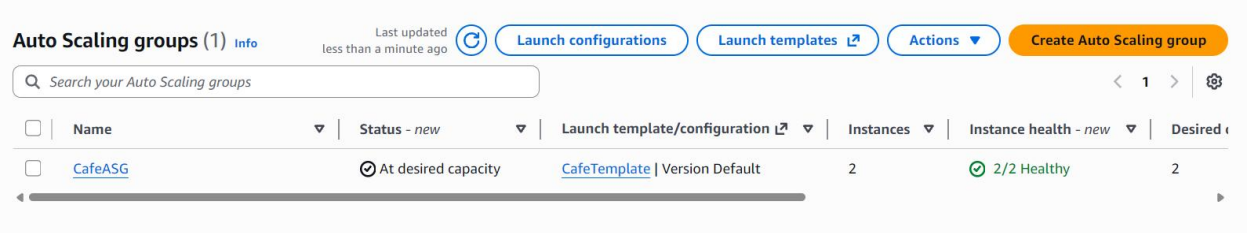
- AMI: Cafe WebServer Image
- Instance type: t2.micro
- Security group: CafeSG
- IAM instance profile: CafeRole (visible in Advanced details)
- Resource tag: Name = webserver



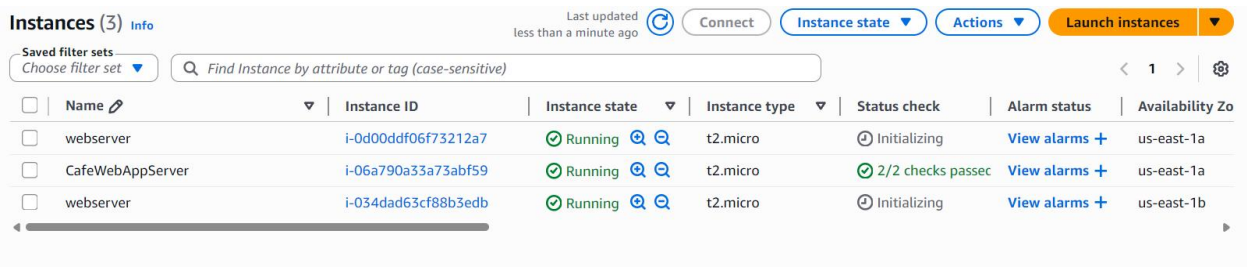
Task 4: Creating an Auto Scaling group

Show two screenshots:

- The Auto Scaling group details page showing: the launch template selected, VPC/subnets (Private Subnet 1 and Private Subnet 2), and group size (Desired: 2, Min: 2, Max: 6).



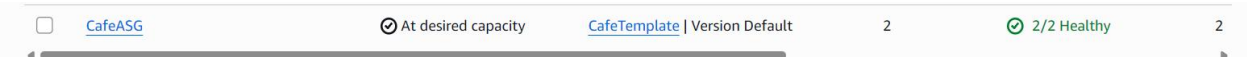
- The EC2 Instances console listing the two running webserver instances launched by the Auto Scaling group, each tagged with the name **webserver**.



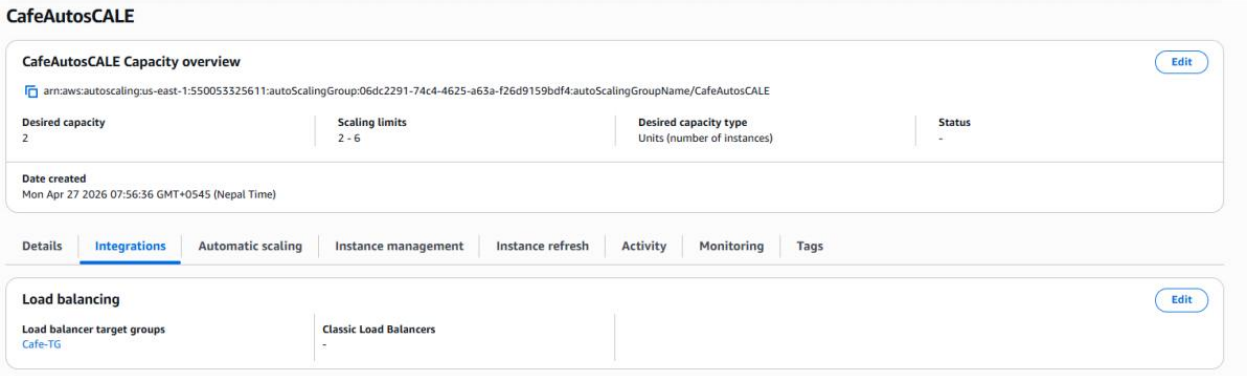
Task 5: Creating a load balancer

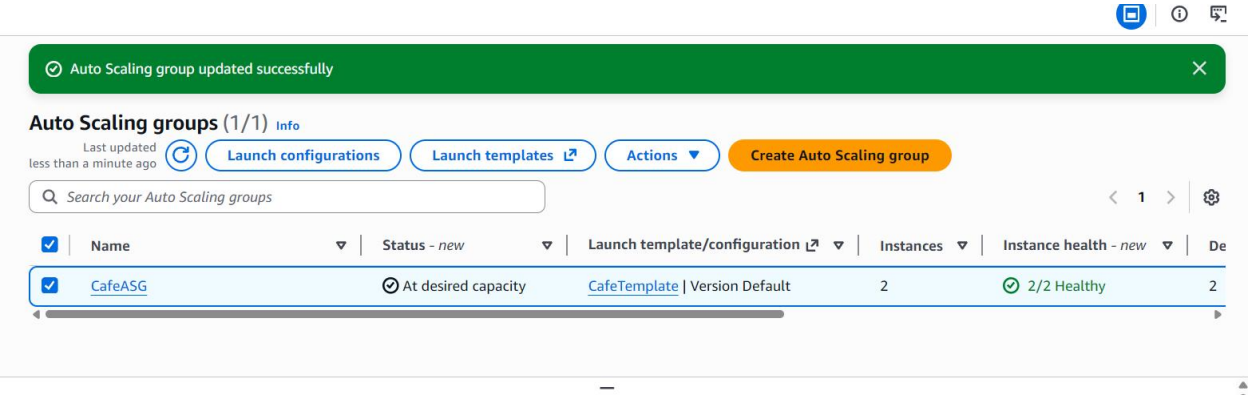
Show two screenshots:

- The EC2 Load Balancers console showing the Application Load Balancer in Active state, with its DNS name visible, placed in the two public subnets.



- The Auto Scaling group's Load balancing section (under Details tab) confirming the target group has been attached to the group.

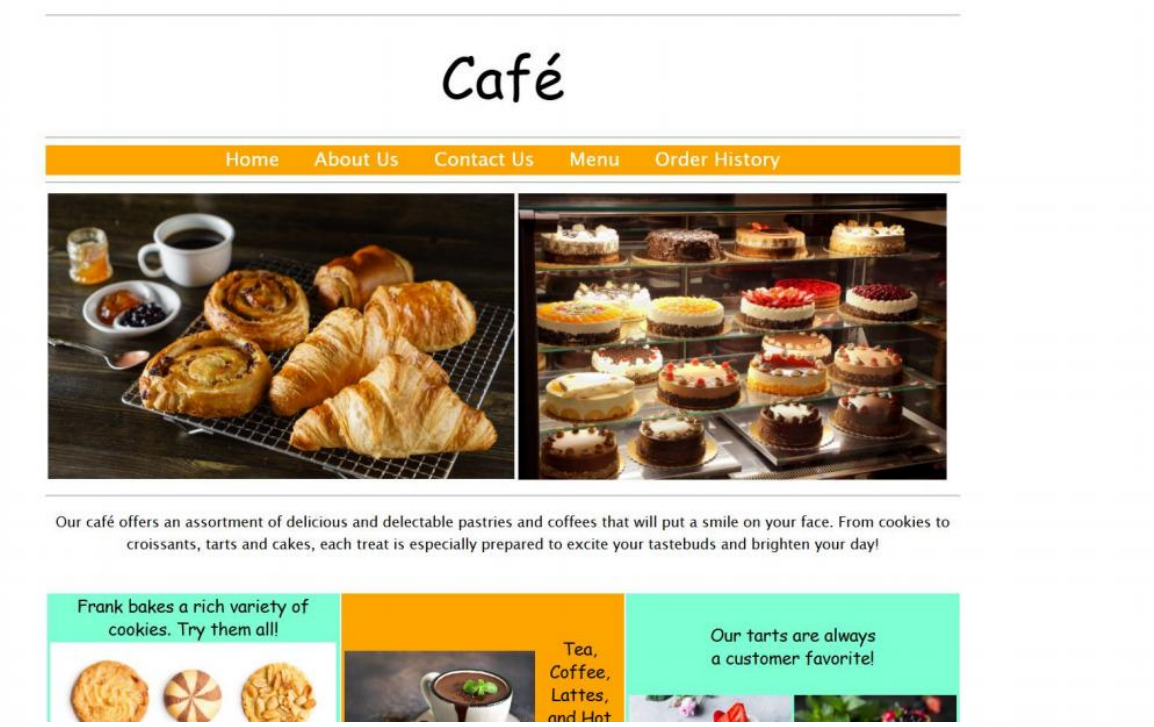




Task 6: Testing load balancing and automatic scaling

Task 6.1: Testing the web application without load

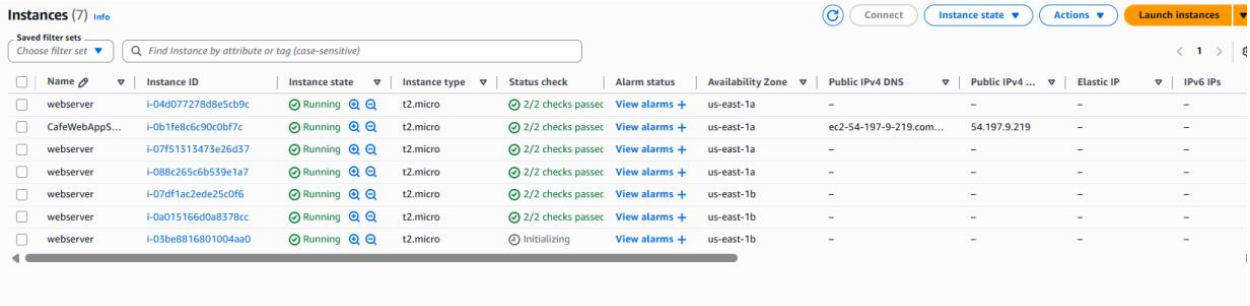
- Show a screenshot of the browser successfully loading the café application via the load balancer DNS name (http://<load-balancer-dns>/cafe), with the full café menu visible.



Task 6.2: Testing automatic scaling under load

Show two screenshots taken after running the stress test (stress --cpu 1 --timeout 600) on one of the web server instances:

- The EC2 console Instances list showing more than two webserver instances in the running state (proof that the Auto Scaling group scaled out in response to the load).
- The Auto Scaling group Activity tab showing the scale-out activity history entry, confirming new instances were launched due to the CPU utilization target tracking policy being triggered.



Instances (7) Info

Connect | Instance state | Actions | Launch instances

Saved filter sets — Choose filter set — Find instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs
webserver	i-04d077278d8e5cb9c	Running	t2.micro	2/2 checks passed	View alarms	us-east-1a	–	–	–	–
CafeWebAppS...	i-0b1fe8c6c90c0bf7c	Running	t2.micro	2/2 checks passed	View alarms	us-east-1a	ec2-54-197-9-219.com...	54.197.9.219	–	–
webserver	i-07f51313473e26d37	Running	t2.micro	2/2 checks passed	View alarms	us-east-1a	–	–	–	–
webserver	i-088c265c6b539e1a7	Running	t2.micro	2/2 checks passed	View alarms	us-east-1a	–	–	–	–
webserver	i-07df1ac2ede25c0f6	Running	t2.micro	2/2 checks passed	View alarms	us-east-1b	–	–	–	–
webserver	i-0a015166d0a8378cc	Running	t2.micro	2/2 checks passed	View alarms	us-east-1b	–	–	–	–
webserver	i-03be8816801004aa0	Running	t2.micro	Initializing	View alarms	us-east-1b	–	–	–	–

No notifications are currently specified

Create notification

Activity history (3)

Filter activity history — Filter by a date and time range

Status	Description	Cause	Start time	End time
Successful	Updating load balancers/target groups: Successful. Status Reason: Added: arn:aws:elasticloadbalancing:us-east-1:550053325611:targetgroup/Cafe-TG/9eabb3de6fb2fa58 (Target Group).		2026 April 27, 08:13:24 AM +05:45	2026 April 27, 08:13:25 AM +05:45
Successful	Launching a new EC2 instance: i-07df1ac2ede25c0f6	At 2026-04-27T02:11:36Z a user request created an AutoScalingGroup changing the desired capacity from 0 to 2. At 2026-04-27T02:11:37Z an instance was started in response to a difference between desired and actual capacity, increasing the capacity from 0 to 2.	2026 April 27, 07:56:39 AM +05:45	2026 April 27, 07:57:11 AM +05:45
Successful	Launching a new EC2 instance: i-04d077278d8e5cb9c	At 2026-04-27T02:11:36Z a user request created an AutoScalingGroup changing the desired capacity from 0 to 2. At 2026-04-27T02:11:37Z an instance was started in response to a difference between desired and actual capacity, increasing the capacity from 0 to 2.	2026 April 27, 07:56:39 AM +05:45	2026 April 27, 07:57:11 AM +05:45

Description

Write in your own wording and clearly cover all of the following (about 250 to 450 words):

1. what you built and what you did in that lab
2. one security control you used and why it matters
3. one performance or reliability feature that is present and how it helps

4. one issue you had and how you fixed it (or one likely issue and how you would fix it)

Description

A highly accessible and scalable web environment was set up for a cafe in anticipation of increased website traffic for this lab. To make sure the VPC and Security Groups were set up correctly, the initial network configuration was audited. Next, a NAT Gateway was provisioned to the public subnet and the route tables for the

In order to enable safe communication between internal servers and the internet for updates, private subnets were modified.
A Launch template was made using "Cafe WebServer Image," and the Auto Scaling group was started from it. Lastly, it was linked to an Application Load Balancer.This ensures that if one server/data center goes
down the website is still available, and when the website receives more traffic than normal the system automatically adds more servers to balance the increased load.

The deployment of the web servers on private subnets was one security measure used in this experiment. This is crucial since the servers cannot be directly accessed from the internet; instead, the Load Balancer filters all requests, effectively protecting the servers. Web servers on private subnets could still access the internet for security updates and other software thanks to the NAT Gateway.
The Auto Scaling and Application Load Balancer configuration's main dependability feature. The load balancer's continuous monitoring guarantees that only operational servers receive the traffic.

During the stress test the CPU utilization on the servers was observed and as one server was intentionally brought to a high usage, new instances were launched and sent online to share the load. This design protects the website from crashing and provides every customer to experience a smooth and well-performing site.

The inability to load the website using the Load Balancer DNS name is one potential issue. I ensured that the load balancer was situated in the public subnets rather than the private ones in order to resolve this. An LB cannot be accessed via the internet if it is located inside a private subnet. In order to access it via the LB, it was also ensured that the "CafeSG" security group has HTTP on Port 80. To maintain the site's dependability, the Auto Scaling group was configured to have at least two servers in different Availability zones.